

Telecom Customer Churn Analysis

Executive Summary Report

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1. Overview

This report presents the findings of a comprehensive exploratory data analysis (EDA) conducted on a telecom customer dataset to understand the key drivers of customer churn. The objective was to identify patterns, segment at-risk customers, and provide actionable recommendations to improve retention.

2. Key Performance Indicators

7,043 Total Customers	1,869 Churned Customers	5,174 Retained Customers
26.54% Churn Rate	73.46% Retention Rate	32.37 months Avg. Tenure
\$64.76 Avg. Monthly Charges	\$2,279 Avg. Total Charges	

3. Key Findings

3.1 Churn Distribution

26.54% of the total customer base has churned, representing 1,869 out of 7,043 customers. This is a significant proportion and warrants urgent attention.

3.2 Gender

Churn rates are nearly identical across male and female customers. Gender is not a meaningful predictor of churn and should not be prioritized in retention strategies.

3.3 Senior Citizens

Senior citizens exhibit a noticeably higher churn rate compared to non-senior customers. This demographic group may require dedicated support programs, simplified service packages, and more proactive outreach.

3.4 Tenure

Tenure is one of the strongest indicators of churn. Customers in their first 1–2 months are at the highest risk of leaving. Customers who have stayed beyond 12 months show significantly stronger loyalty. This pattern underscores the critical importance of the onboarding experience.

3.5 Contract Type

Month-to-month contract holders churn at a significantly higher rate than customers on 1-year or 2-year contracts. Long-term contracts act as a natural retention mechanism.

3.6 Services Subscribed

Customers without add-on services such as Online Security, Tech Support, Online Backup, and Device Protection are far more likely to churn. Customers engaged with more services demonstrate stronger retention.

3.7 Payment Method

Customers paying via Electronic Check have the highest churn rate compared to those using credit card, bank transfer, or mailed check. This may indicate a lower level of engagement or commitment to the service.

3.8 Internet Service

Fiber Optic internet users churn at a higher rate than DSL users, possibly reflecting unmet performance expectations or pricing concerns at higher service tiers.

4. Findings at a Glance

Factor	Finding	Risk Level
Gender	No significant difference in churn	Low
Senior Citizen	Higher churn among seniors	High
Tenure (< 2 months)	Highest churn concentration	Critical
Month-to-Month Contract	Most prone to churn	Critical
No Online Security	Strong churn predictor	High

No Tech Support	Strong churn predictor	High
Electronic Check Payment	Highest churn by payment method	High
Fiber Optic Internet	Higher churn vs. DSL	Medium

5. Recommendations

01

Strengthen the Onboarding Experience

New customers (0–3 months) are at the highest risk of leaving. Implement a structured welcome program, regular check-ins, and usage guidance in the first 90 days.

02

Incentivize Long-Term Contracts

Offer loyalty discounts, bundled benefits, or service credits to encourage customers on month-to-month plans to upgrade to annual or biannual contracts.

03

Bundle Value-Added Services

Promote bundled packages that include Online Security, Tech Support, and Device Protection. Customers with more services are significantly less likely to churn.

04

Senior Citizen Retention Program

Design simplified, dedicated service plans for senior customers. Consider dedicated support lines and simplified billing options.

05

Payment Method Migration

Run campaigns to migrate electronic check users to auto-pay via credit card or bank transfer, which correlates with lower churn rates.

06

Review Fiber Optic Value Proposition

Investigate whether Fiber Optic pricing, reliability, or service quality is meeting customer expectations and address pain points proactively.

6. Conclusion

This analysis reveals that telecom customer churn is not random — it is driven by identifiable, addressable factors. The highest-risk customers are those who are new to the service, on flexible month-to-month contracts, using Electronic Check payments, and not subscribed to value-added services.

By investing in early engagement, long-term contract incentives, service bundling, and targeted outreach for senior citizens, the company has a clear pathway to significantly reduce its 26.54% churn rate and improve overall customer lifetime value.

Prepared using Python (Pandas, Seaborn, Matplotlib) • Dataset: Customer Churn.csv